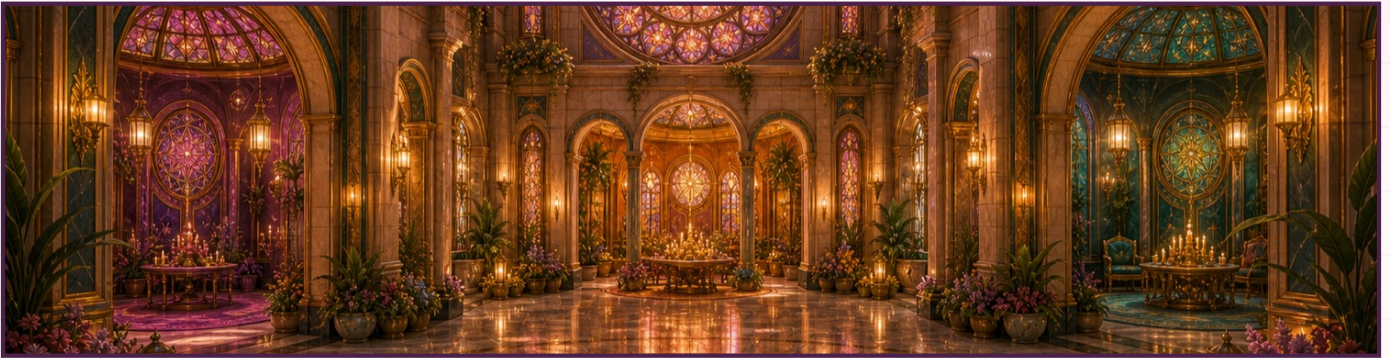


The Founding Mothers

Almost two hundred years, and every real leap has a woman's name on it. Whether or not the textbook wrote it down.



THE QUIET WING AT THE BACK OF THE LUMINAIRY

"You were never behind on AI. You were just never told it was yours - built by women, for almost two hundred years, until it finally knocked on your door."

PIN THIS UP

What's inside: the whole family tree of the machine on your desk (page 2), the four Study Pack words fully defined (page 3), and this week's field trip plus the happy-hour answer (page 4).

Stick it where the meeting can see it. The next time someone says the science is new, you have receipts.

SAY THEIR NAMES

Ada Lovelace the idea, 1843

Hedy Lamarr the signal, 1942

The ENIAC Six the first programmers, 1945

Grace Hopper the language, 1952

Karen Spärck Jones the finding, 1972

Fei-Fei Li the sight, 2012

Joy · Timnit · Emily · Kate the checkers, now

How we got here

1843 Ada Lovelace • The Idea

Everyone saw an adding machine. She saw that if a machine can follow instructions, *numbers are only the beginning* - and wrote the first algorithm. The credit went to the man whose machine it was.

1942 Hedy Lamarr • The Signal

The most beautiful woman in the world, inventing between takes: a signal that hops frequencies too fast to jam. It's in the family tree of your Wi-Fi and Bluetooth. She never made a dime.

1945 The ENIAC Six • The First Programmers

Six women programmed the first general-purpose electronic computer by hand, before programming languages existed. They're in the famous photos - uncaptioned for about forty years.

1952 Grace Hopper • The Language

Told a computer could never understand words, she called that *a failure of imagination* and built the compiler - so we talk to it in our words. Also taped the first actual bug into a logbook.

1956 Dartmouth • The Naming (then the winters)

A summer room of men christens it "artificial intelligence" and promises the world. The promises outran the results, the money froze - roughly two AI winters. The work went quiet, not away.

1972 Karen Spärck Jones • The Finding

The rare words carry the meaning - the arithmetic under every search box since. Her line: *"Computing is too important to be left to men."*

2012 Fei-Fei Li • The Sight

The brain was never the problem - the machine had just never seen enough examples. Her millions of labeled images (training data) started the boom. The Godmother of AI.

2022 It lands on your desk

One ordinary Wednesday in November, all of it gets wrapped in a little chat box and put online for free. About 100 million people in two months. The science is old; your access is brand new.

Now Joy Buolamwini • Timnit Gebru • Emily Bender • Kate Crawford

The mirror, the room, the parrot, the map - the women who check the whole machine, so the thing you trust is worth trusting.

Four words worth keeping

ADA, 1843

Algorithm

A precise, repeatable set of steps for solving a problem - the recipe a machine follows, in order, every time. Ada Lovelace wrote the first one in 1843, for a machine that wouldn't be built for another century. Not magic, not thinking: just very careful instructions, followed exactly.

GRACE, 1952

Compiler

A translator that turns instructions written in something close to plain language into the raw code a machine actually runs. Grace Hopper built one of the first, so people would stop having to think like the machine and could tell it what they wanted in words. Every app you tap sits on top of that one idea.

ROUGHLY TWO OF THEM

AI winter

One of the long stretches - there were roughly two (about 1974-1980, and 1987-1993, with an "expert systems" boom between them) - when AI's big promises outran what it could actually do, the funding froze, and "artificial intelligence" became a slightly embarrassing thing to say out loud. The work didn't stop; it just went quiet, until the thaw.

FEI-FEI, 2012

Training data

The examples a model learns from - the more it sees, the more it can do. Fei-Fei Li's insight was that the bottleneck was never the machine's "brain"; it was that no one had shown it enough of the world. Her ImageNet - millions of labeled pictures - is why, in 2012, a machine finally learned to see.

Each of these four deserves more than a card. Full deep-dives are in the works at the LIBRAiRY.

No homework. A field trip.

Go meet a MAiVEN

1. Go to the LUMINAiRY and walk into the quiet wing at the back - the MAiVENS.
2. Pick one woman you'd genuinely never heard of before today. Read what she did and what it cost her.
3. Text one friend a single sentence about her. One name, one sentence, one friend. That's the whole assignment.

SAY IT AT HAPPY HOUR

"Is AI brand new? It's almost two hundred years old and about three years old at the same time. The science has been building for centuries - your access to it is brand new. And every real leap in it has a woman's name on it."

SO REMEMBER, LADIES

"You were never behind on AI. You were just never told it was yours - built by women, for almost two hundred years, until it finally knocked on your door."